

Dataset-specific handling

Vrbancic et al. (2020)

88,647 rows | 111 features
34.57% phishing (1:1.89)

Target column: phishing
all features already numeric

URL-Phish (2025)

116,600 rows | 22 features
14.24% phishing (1:6.02)

Target column: label
drop url, dom, tld identifiers

Applied identically to both datasets

Map labels to a single convention

phishing = positive class (1), legitimate = 0

Structural cleaning

remove zero-variance and duplicated columns
replace non-finite values (neither dataset has missing values)

Stratified subsample to 20,000 rows

proportional within each class, so the natural ratio is preserved
Vrbancic 34.57% | URL-Phish 14.23% after reduction
volume reduced only, to keep SVM training tractable

Stratified 80/20 train-test split

class proportion preserved in both partitions
seed controls the draw (42, 1 or 2)

Training set (16,000)

passed to imbalance treatment
and cross-validation

Inside each CV fold

1. scaler fitted on fold training part
2. resampling applied to it only
3. classifier trained

Test set (4,000)

sealed; no resampling
and no scaling fitted on it

*held back until
final evaluation*

Final evaluation on the untouched test set